Congenital Diaphragmatic Hernia: Survival Treated With Very Delayed Surgery, Spontaneous Respiration, and No Chest Tube

By J.T. Wung, R. Sahni, S.T. Moffitt, E. Lipsitz, and C.J.H. Stolar
New York, New York

● This report suggests that stabilization of the intrauterine to extrauterine transitional circulation combined with a respiratory care strategy that avoids pulmonary overdistension, takes advantage of inherent biological cardiorespiratory mechanics, and very delayed surgery for congenital diaphragmatic hernia results in improved survival and decreases the need for extracorporeal membrane oxygenation (ECMO). This retrospective review of a 10-year experience in which the respiratory care strategy, ECMO availability, and technique of surgical repair remained essentially constant describes the evolution of this method of management of congenital diaphragmatic hernia.

INDEX WORDS: Congenital diaphragmatic hernia; persistent pulmonary hypertension; extracorporeal membrane oxygenation.

SINCE THE ADVENT of extracorporeal membrane oxygenation (ECMO), infants with overwhelming cardiorespiratory failure,[1] who would have otherwise died, now survive. The ability of ECMO to salvage a significant number of infants with congenital diaphragmatic hernia (CDH) has stimulated the thought process regarding optimal care for these infants.[2] The concept of ECMO as a safety net has allowed the contemporary clinician to re-evaluate traditional dogma and test new approaches of management. The prior aphorism that CDH is a surgical emergency has been successfully challenged.[3-10] Although CDH is a complex interface of pulmonary hypoplasia and pulmonary hypertension, disturbances of lung development occur long before birth and are unlikely to be acutely modified by emergent surgical repair.[11-14] The clinical situation may be aggravated by surfactant deficiency[15] and left ventricular hypoplasia.[16] As a result, severely affected infants with CDH have difficulty adjusting from intrauterine to extrauterine life and need additional time for vascular remodeling to make the adjustment. Body fluid shifts and reduced pulmonary compliance are factors that may also adversely affect the CDH patient after emergent surgical repair.[17]

The life-threatening and nonhomogenous nature of severely affected infants with CDH makes strict randomized treatment programs difficult if not virtually impossible to carry out. However, new strategies evolved with time and experience, leading us to treat the disturbed physiology of patients with CDH as an emergent problem rather than the disturbed anatomy. Recognizing that preductal arterial oxygen content may be a good clinical marker of the neonatal lung's potential for meaningful gas exchange[18] has allowed us to pursue alternate respiratory care strategies that permit adequate oxygen delivery and avoid unwarranted clinical decisions based on postductal blood gas analysis, and furthermore, minimize ill-advised use of ECMO in infants with pulmonary hypoplasia. Overdistention of the lungs in babies with CDH can exacerbate arteriolar vascular resistance, causing pulmonary hypertension, and also result in lung injury caused by barotrauma.[19,20] We have developed a respiratory care strategy that maintains adequate oxygen delivery, minimizes iatrogenic injury and exacerbation of pulmonary hypertension by avoiding overventilation and alkalinization, and takes advantage of the mechanics of spontaneous respiration.

Although this is a retrospective nonrandomized study, the basic respiratory care strategy, surgical management, and indications for ECMO in CDH remained essentially similar for the duration of the study. This article reviews our experience with CDH and allowed us to evaluate (1) whether delayed operative repair was possible, (2) whether delaying the surgical procedure until there was no preductal or postductal oxygen saturation gradient with minimal supplemental oxygen was possible, (3) whether avoiding the use of chest tubes to reduce the risk of wide swings in intrathoracic pressure was feasible, and (4) whether these concepts of management influenced survival or the need for ECMO.

MATERIALS AND METHODS

All infants reviewed were treated at a single institution between December 1983 and December 1995 by the same group of clinicians. They were either diagnosed inborn at the time of birth,

From the Divisions of Pediatric Surgery and Neonatology, Columbia University, College of Physicians and Surgeons, and Babies and Children's Hospital of New York, Columbia-Presbyterian Medical Center, New York, NY.
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Supported by The Anya Fund and The Charles Edison Fund.
Address reprint requests to Charles J.H. Stolar, MD, Babies' and Children's Hospital of New York, Room 212 North, 3959 Broadway, New York, NY 10032.

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prenatally diagnosed inborn, or transferred preoperatively from another institution at less than 4 hours of life. All reported patients required intubation for respiratory distress at less than 2 hours of life. Severity of respiratory distress was estimated by A-aDO_2 calculation [A-aDO_2 = [P(barometric)-47-$PaCO_2$-PaO_2]. Infants transferred after surgery or not requiring mechanical ventilation at birth were excluded.

Respiratory Care Strategy

All infants were nasally intubated and ventilated with a time-cycled, pressure-limited, and continuous-flow infant ventilator (Infant Star 200, Infrasonics Inc, San Diego, CA). Ventilator strategy followed an established consistent protocol. Infants were ventilated initially in a conventional mode with an intermittent mandatory rate (IMV) of 20 to 40 breaths/min, peak inspiratory pressure (PIP) of 20 to 25 cm H_2O for adequate chest and/or abdominal excursions, and positive end expiratory pressure (PEEP) of 5 cm H_2O. Sufficient supplemental oxygen was administered to maintain preduct $SaO_2 > 90\%$. If oxygenation was not adequate, very labile, or $PaCO_2 > 60$ mm Hg, an unconventional or high-frequency positive-pressure ventilation (HIPPV) mode was used. HIPPV used the same ventilator, but with IMV of 100 breaths/min, PIP of 20 cm H_2O, and PEEP of 0 cm H_2O. Respirator settings were weaned as soon as pulmonary status improved. The goal of the strategy was to maintain preductal $SaO_2 > 80\%$ with minimal but adequate settings, without muscle relaxants except during surgery, and with avoidance of hyperventilation or other attempts to induce alkalosis. Hyperinflation of these vulnerable lungs was assiduously avoided both preoperatively and postoperatively. All infants were allowed to breathe spontaneously. Narcotics for analgesia were administered as needed. The extent of respiratory support depended on the degree of respiratory failure and was minimized to control the risk of barotrauma and cardiovascular compromise. If the ventilator settings were adequate to move the chest wall but hypoxemia persisted because of pulmonary hypertension, vasodilator therapy (tolazoline, 1 to 2 mg/kg; or dolbutamine, 5 to 10 μg/kg/h) was used instead of increasing PIP or IMV.

Timing of Surgery

Over the 10 years of this study, infants fell into one of three groups as the strategy evolved. The earliest patients, group 1 (n = 17), underwent emergency surgery. Repair of the CDH was accomplished as rapidly as possible, often with no preoperative care in the neonatal intensive care unit, or immediately after elective cesarean section in the cases of prenatal diagnosis. Group 2 (n = 28) had a modest delay in surgery arbitrarily determined to be approximately 24 hours of limited initial stabilization. All infants in groups 1 and 2 had prophylactic chest tubes that were placed in the ipsilateral hemithorax at the time of hernia repair and that were attached to 2 to 5 cm H_2O seal with no suction. Group 3 (n = 18) infants had very delayed surgery until there was no preductal or postductal SaO_2 gradient after weaning ventilator support and supplemental oxygen, and the echocardiographic examination showed minimal evidence of pulmonary hypertension. Group 3 did not undergo surgery until ventilator support was weaned to a minimum and had been stable for 24 hours. Weaning of ventilator support was guided by preductal oximetry and physical examination. No patient in group 3 had a chest tube placed for any reason.

Surgery

All diaphragm repairs were accomplished through an abdominal approach. A subcostal exposure allowed reduction of abdominal viscera from the ipsilateral thorax. Primary repair of the defect was done only if feasible with minimal dissection. If not, a 1-mm Gore-tex (Gore Association, Tuscon, AZ) prosthesis was used and fixed in an anatomic position with interrupted nonabsorbable sutures. Special adjuncts including soft tissue flaps, abdominal wall stretching, manual extrusion of meconium through the anus or correction of malrotation with appendectomy were not performed. If the abdominal wall could not be closed without tension, another Gore-tex prosthesis was used. The infants with abdominal wall prosthesis underwent serial reduction and subsequent primary abdominal wall closure after decannulation from ECMO. Calcium-activated fibrinogen was applied to cut surfaces for hemostasis. No ϵ-amino caproic acid was used.

ECMO Indications

ECMO was considered appropriate for an infant in whom conventional therapy failed only if a preductal SaO_2 of at least 95% or a measured PO_2 of at least 100 mm Hg with a PCO_2 less than 50 mm Hg was observed at any time in the pre-ECMO course. ECMO was also considered if the infant subsequently showed evidence of inadequate oxygen delivery, progressive metabolic acidosis, and organ failure despite maximal conventional therapy. Extracorporeal support was instituted before inordinate lung distending pressures were used. An A-a$DO_2 > 600$ mm Hg for more than 4 hours was generally considered an indication for ECMO. ECMO was not considered without some evidence of potentially adequate lung parenchyma. ECMO was instituted preoperatively for a patient unable to be stabilized only if there was reasonable expectation of adequate lung parenchyma and potentially reversible pathophysiology.

ECMO Technique

ECMO was venoarterial in all cases. The extrathoracic cannulation was accomplished at the bedside using a 10 F end-hole arterial perfusion cannula and a 12 F multiple side-hole venous drainage catheter (Elecath, Rahway, NJ). Flows were sufficient to meet tissue oxygen requirements with minimal ventilator and supplemental oxygen support, no more than 125 mL/kg/min. Weaning from ECMO was guided primarily by mixed venous oxygen saturation that was maintained at 65% to 70%. ECMO support was carried out by standard techniques previously published.[1] Patients placed on ECMO preoperatively underwent CDH repair when ECMO had been maximally reduced, and treatment was discontinued as soon as possible after the repair.

Data Analysis

All preoperative infants with immediate respiratory distress requiring mechanical ventilation only were included. Outcomes were analyzed by χ^2 test, paired t test, or analysis of variance (ANOVA) and expressed as mean ± SD.

RESULTS

Seventy-one infants with CDH evaluated preoperatively were included. Four infants with multiple anomalies and 4 infants with pulmonary hypoplasia (determined by preductal blood gas evaluation) were excluded. The 4 infants excluded for pulmonary hypoplasia had <5% tile for lung weights at autopsy (group 1, 1; group 2, 2; group 3, 1). Sixty-three remaining infants fell into one of the three study groups.

Table 1. Description and Outcome for Groups 1, 2, and 3

	Group 1	Group 2	Group 3	P
Number of infants	17	28	18	NS
Weight	3.0 ± 0.5	3.1 ± 0.6	3.1 ± 0.4	NS
Age at surgery (h)	5.9 ± 5.8	21.6 ± 23.0	100.1 ± 44	$P < 0.001$
Postoperative IMV(d)	5.3 ± 3.9	5.6 ± 4.0	3.6 ± 2.6	NS
Survival (overall)	14(82%)	21(75%)	17(94%)	$P < .04$ versus groups 1 & 2
Treated with ECMO	6/17(35%)	7/28(25%)	1/18(6%)	$P < .03$ versus groups 1 & 2
ECMO contraindicated	2	3	1	
ECMO survival	5/6	3/7	1/1	

NOTE: Group 3 had a better survival and less need for ECMO compared with groups 1 and 2.

The results of this review are listed in Table 1. Group 1 (immediate surgery and prophylactic chest tube) had 17 infants; group 2 (limited stabilization, surgery within 24 hours, and chest tube), 28; and group 3 (very delayed surgery and no chest tube), 18. The birth weights and gestational ages were the similar in all groups. The time interval until surgical repair was significantly longer for group 2 compared with group 1, group 3 compared with group 1, and group 3 compared with group 2 ($P = .001$). The total number and postoperative number of days of mechanical ventilation were not significantly different between groups. The need for ECMO was different for group 3 compared with groups 1 and 2, but not for group 1 compared with group 2. Although 35% (6/17) of group 1, and 25% (7/28) of group 2 needed ECMO, only 6% (1/18) of group 3 became sufficiently ill to need it ($P < .03$). Six patients in the aggregate group were sufficiently ill but had contraindications to ECMO (intrcranial hemorrhage, 2; pulmonary hemorrhage, 2; seizures, 2). Five of six group 1 patients, 3/7 group 2, and the single group 3 ECMO patient survived and are included in the overall survival. The survival for group 3 was significantly better ($P < .04$) compared with group 1 and group 2. The survival for groups 1 and 2 were not significantly different. In addition, the need for ECMO to accomplish survival was significantly less for group 3 compared with group 1 and 2 ($P < .03$).

DISCUSSION

The notion that CDH might not be a surgical emergency was first proposed by Sakai et al[21] and later confirmed by Nakayama et al.[22] They showed that immediate surgical repair of CDH was followed by a decrease in respiratory compliance. Although Sakai et al hypothesized the compliance changes were caused by aggregate stresses of anatomical repair, the alterations were more likely caused by the usual postoperative alterations in water and sodium balance.[17] Gang et al[23] showed progressive improvement in pulmonary compliance during ECMO in a mixed neonatal ECMO population. Exactly what happens to the cardiorespiratory system during preoperative or ECMO stabilization is unclear. Moffitt et al[24] studied 10 neonates with CDH using continuous, noninvasive, whole-body plethysmography. They found that measures of vascular and pulmonary function undergo essentially the same changes as the normal transitional circulation but over a much more protracted time course. Although unstable initially, these infants had improved gas exchange, compliance, and minute ventilation while the mechanical work of breathing and supplemental oxygen requirement decreased. Deviations from these norms may identify higher-risk patients. In fact, some clinicians believe an infant with CDH unable to be stabilized before surgical repair, regardless of time, has overwhelming pulmonary hypoplasia incompatible with life.[18,25]

The key feature that makes the present report different from other previous studies is the respiratory care strategy designed to minimize overdistension of the lungs.[26,27] In most other reports concerning patients with CDH, treatment includes muscle paralysis, hyperventilation, and induced alkalosis. We believe these techniques serve only to exacerbate pulmonary hypertension and induce barotrauma. A report by Levine et al[20] measured pulmonary vascular resistance under increasing distending pressures with and without hypoxemia. In both conditions, pulmonary vascular resistance was augmented by increased distending pressures. Data reported by Mansell et al[19] concerned the relationship between the small airways and arterioles and showed an increase in arterial resistance as the parenchymal pressure increased and the lung was stretched. The coiled muscular coats around the intra-acinar arterioles may be analogous to a Chinese finger trap. Ventilating at minimal airway pressures not only minimizes barotrauma, but also minimizes interference with systemic venous return. Avoiding muscle paralysis allows the infant's own respiratory bellows to contribute to minute ventilation without compromising functional residual capacity.

Unless there is active bleeding or other sources for hemothorax or pneumothorax, a chest tube is not needed. The ipsilateral lung is smaller than its hemithorax, and the pleural space seeks to be obliterated. Because the ipsilateral lung is unable to fill the space immediately, it will passively fill with fluid. A chest tube prolongs this process, but also, by allowing the mediastinum to swing to the ipsilateral side, causes undue stretch and distension of the contralateral side with the above-noted consequences.

In the infant with CDH, a strategy based on minimal iatrogenic ventilator injury, avoiding overdistension of these brittle lungs, and careful attention to oxygen delivery as measured by preductal assesment, will allow the transitional circulatory changes that ordinarily accompany birth to take places during a prolonged preoperative period. Survival should be improved and the need for ECMO minimized.

REFERENCES

1. Stolar CJH, Snedecor SM, Bartlett RH: Extracorporeal membrane oxygenation for neonatal respiratory failure. J Pediatr Surg 26:563-571, 1991
2. Weinstein S, Stolar CJH: Newborn surgical emergencies—Congenital diaphragmatic hernia and extracorporeal membrane oxygenation. Pediatr Clin North Am, 40:1315-1334, 1994
3. Heiss K, Manning P, Oldham KT, et al: Reversal of mortality for congenital diaphragmatic hernia with ECMO. Ann Surg 209: 1225-1230, 1989
4. West KW, Bengston K, Rescorla FJ, et al: Delayed surgical repair and ECMO improves survival in congenital diaphragmatic hernia. Ann Surg 216:454-462, 1992
5. Breux CW, Rouse TM, Cain WS, et al: Improvement in survival of patients with congenital diaphragmatic hernia using a strategy of delayed repair after medical and/or extracorporeal membrane oxygenation stabilization. J Pediatr Surg 26:333-338, 1991
6. Langer JC, Filler RM, Bohn DJ, et al: Timing of surgery for congenital diaphragmatic hernia: Is emergency surgery necessary? J Pediatr Surg 23:731-734, 1988
7. Shanbhogue LKR, Tam PKH, Ninan G, et al: Preoperative stabilisation in congenital diaphragmatic hernia. Arch Dis Child 65:1043-1044, 1990
8. Coughlin JP, Drucker DEM, Cullen ML, et al: Delayed repair of congenital diaphragmatic hernia. Am Surg 55:90-93, 1993
9. Wilson JM, Lund DP, Lillehei CW, et al: Delayed repair and preoperative ECMO does not improve survival in high risk congenital diaphragmatic hernia. J Pediatr Surg 27:268-375, 1992
10. Lally KP, Paranka MS, Roden J, et al: Congenital diaphragmatic hernia: Stabilization and repair on ECMO. Ann Surg 216:569-573, 1992
11. Geggel RL, Murphy JD, Langleben D, et al: Congenital diaphragmatic hernia: Arterial structural changes and persistent pulmonary hypertension after surgical repair. J Pediatr 107:457-464, 1985
12. Hislop A, Reid L: Pulmonary arterial development during childhood: Branching pattern and structure. Thorax 28:129-135, 1973
13. Kitagawa M, Hislop A, Boyden EA, et al: Lung hypoplasia in congenital diaphragmatic hernia: A quantitative study of airway, artery, and alveolar development. Br J Surg 58:342-356, 1971
14. Levin DL: Morphologic analysis of the pulmonary vascular bed in congenital left sided diaphragmatic hernia. J Pediatr 92:805-809, 1978
15. Karamanoukian HL, Glick PL, Wilcox DT, et al: Congenital diaphragmatic hernia is deficient in surfactant. J Pediatr Surg (in press)
16. Beals DW, Wilson JW, Reid L: Left ventricular hypoplasia in severely affected infants with congenital diaphragmatic hernia. J Pediatr Surg (in press)
17. Moore FD, Oleson KH, McMurey JD, et al (eds): Body cell mass and its supporting environment, in Body Composition in Health and Disease. Philadelphia, PA, Saunders, 1973
18. Stolar CJH, Dillon PW, Reyes C, et al: Selective use of extracorporeal membrane oxygenation in the management of congenital diaphragmatic hernia. J Pediatr Surg 23:207-211, 1988
19. Mansell AL, McAteer AL, Pipkin AC: Maturation of interdependence of extra-alveolar arterioles and lung parenchyma in piglets. Circ Res 71:701-710, 1992
20. Levine G, Boyd G, Milstein J, et al: Influence of airway state on hemodynamics of the pulmonary circulation in newborn lambs. Pediatr Res 31:1869, 1992
21. Sakai H, Tamura M, Hosokowa Y, et al: Effect of surgical repair on respiratory mechanics in congenital diaphragmatic hernia. J Pediatr 111:432-438, 1987
22. Nakayama DK, Motoyama EK, Tagge EM: Effect of preoperative stabilization on respiratory system compliance and outcome in newborn infants with congenital diaphragmatic hernia. J Pediatr 118:793-799, 1991
23. Gang M, Lew CD, Ramos AM, et al: Serial measurements of pulmonary mechanics assists in weaning from extracorporeal membrane oxygenation in neonates with respiratory failure. Chest 100:771-773, 1991
24. Moffitt ST, Schulze KF, Sahni R, et al: Preoperative cardiorespiratory trends in infants with congenital diaphragmatic hernia. J Pediatr Surg (in press)
25. Bohn D, Tamura M, Perrin D, et al: Ventilatory predictors of pulmonary hypoplasia in congenital diaphragmatic hernia confirmed by morphologic assesment. J Pediatr 111:423-431, 1987
26. Wung J-T: Mechanical ventilation using conventional infant respirators, in Pomerance JJ, Richardson CJ (eds): Neonatology for the Clinician. Pomerance Norwalk, CT, Appleton & Lange, 1993, pp 279-287 (chap 23)
27. Wung J-T, James LS, Kilchevsky E, et al: Management of infants with severe respiratory failure and persistance of the fetal circulation without hyperventilation. Pediatrics 76:488-494, 1985